



MASTER CODE

Project Name: Café Recommendation System

Group Members:

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1. Introduction

- C++ console-based program simulating a smart café.
- Demonstrates arrays, functions, loops, conditionals, and formatted input/output.
- Allows menu display, order placement, AI-based suggestions, and bill generation.

2. Objectives

- Display menu with Veg/Non-Veg and Spicy/Non-Spicy items.
- Suggest items based on budget and customer preferences.
- Take orders with quantities and generate detailed bills including GST.
- Improve programming skills with practical application.

3. Key Features

- Menu Display: Formatted table using arrays and setw.
- AI Recommendations: Suggest items within budget and preference constraints.
- Order Placement: Multiple items with quantity input and validation.
- Bill Generation: Subtotal, GST calculation (16% cash, 5% card), total payable.
- Order Completion: Confirmation message simulating café service.

4. Learning Outcomes

- Practical use of arrays, functions, loops, and conditionals.
- Input validation and user interaction.
- Formatted output using <iomanip>.
- Use of different datatypes int, char, bool
- Use void which show returning no value.
- Basic AI-like logic for recommendations.
- Basic bill calculation including tax.

5. Project Description

The Café Recommendation System is a C++ console-based program designed to simulate a smart café ordering system.

The program allows users to:

- View the café menu (Veg / Non-Veg, Spicy / Non-Spicy options)
- Get food suggestions based on their budget and preference
- Select items and enter quantity
- Generate a complete bill
- Calculate GST (16% for cash payment and 5% for card payment)

6. This project demonstrates the practical implementation of:

- Arrays
- Loops
- Conditional statements
- Functions
- Menu-driven programming
- Formatted output using `iomanip`

The system is user-friendly and runs completely in the console.

7. Tools / Technologies Used

- Programming Language: C++
 - Compiler: Dev C++ / CodeBlocks / Any C++ Compiler
 - Libraries Used:
 - `iostream`
 - `iomanip`
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8. Instructions to Run the Project

1. Open the source code file in a C++ compiler (Dev C++, CodeBlocks, etc.).
 2. Compile the program.
 3. Run the program.
 4. Follow the on-screen instructions to:
 - View the menu
 - Enter your preferences
 - Place your order
 - Generate the final bill
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